

Lecture 19:
Policy Lab
Does rural electrification work? II

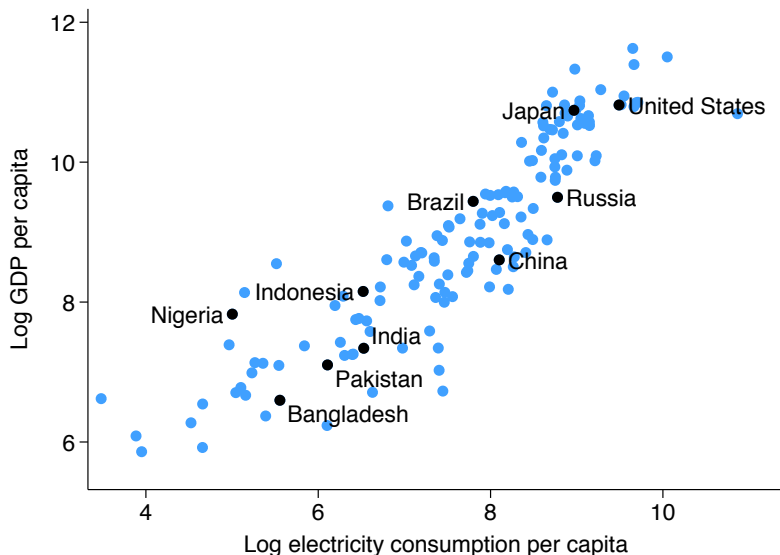
PPHA 34600
Prof. Fiona Burlig

Harris School of Public Policy
University of Chicago

From last time: sorry not sorry



From last time: rural electrification policy lab



Source: World Bank tables

What is the causal effect of rural electrification on economic development?

This is not totally straightforward to answer:

- Naive estimator: compare electrified to non-electrified places
- **Why is this problematic?**
- Electrified places might be...:
 - Growing faster (slower) than non-electrified places
 - More (less) politically connected
 - Have other infrastructure (roads, etc)
 - Be wealthier (less wealthy)
 - Etc

→ There are many forms of selection bias!

From last time: Does rural electrification work (in SA)?

Dinkelman (2011) is a seminal study of rural electrification:

- Estimates effects in post-Apartheid South Africa
- Deals with identification with an instrumental variables approach
- IV: Land gradient
- **Finding:** Rural electrification causes large changes in female employment

Alternative approaches to handling selection bias

We may find the geographic IV unsatisfying...

Today: two additional estimation approaches:

- ① Lee, Miguel, Wolfram (2019): **RCT**
- ② Burlig and Preonas (2016): **RD**

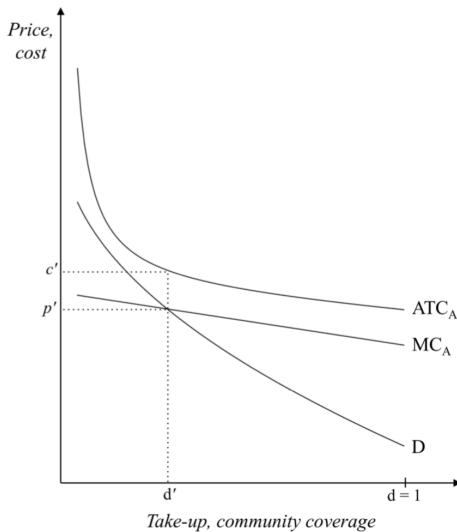
Second paper: Lee, Miguel, Wolfram (2019)

This is the first prominent econ paper to randomize RE:

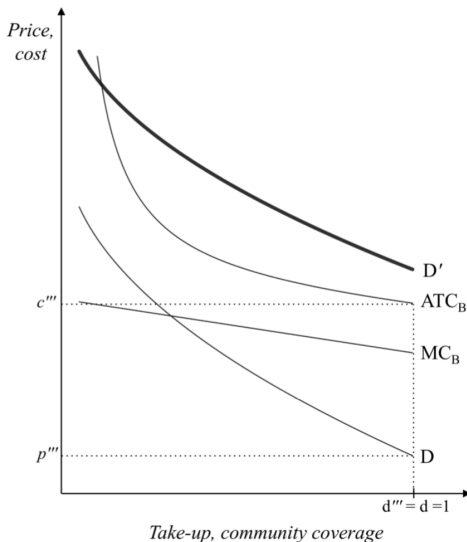
Research question: What are the “economics of rural electrification”?

- AKA, what are the costs and benefits of rural electrification?
- What are the effects of electrification on a variety of “benefits?”
- And how much does it cost to achieve these benefits?

Theoretical framework: the utility as a natural monopoly



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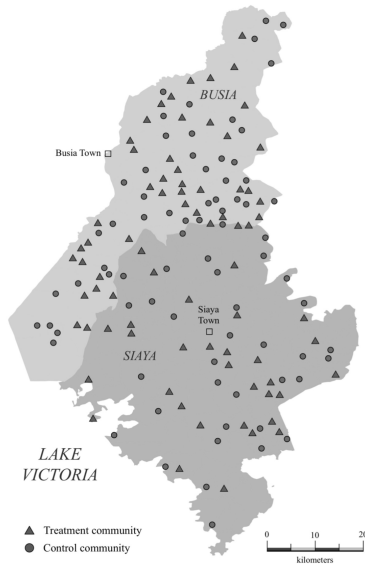
Lee, Miguel, and Wolfram study rural electrification in Kenya:

- Kenya's grid is green(ish): lots of hydro and geothermal
- Lots to do: Installed MW to increase 10 fold by 2031
- Huge increase in electricity access in recent years: REA pushed to electrify public places...
- ... but households remained at low levels (only 32% electrified in 2014)

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- ... but households remained at low levels (only 32% electrified in 2014)
- Households could pay for grid connections: **\$398** within 600m of a transformer

LMW (2019): Context



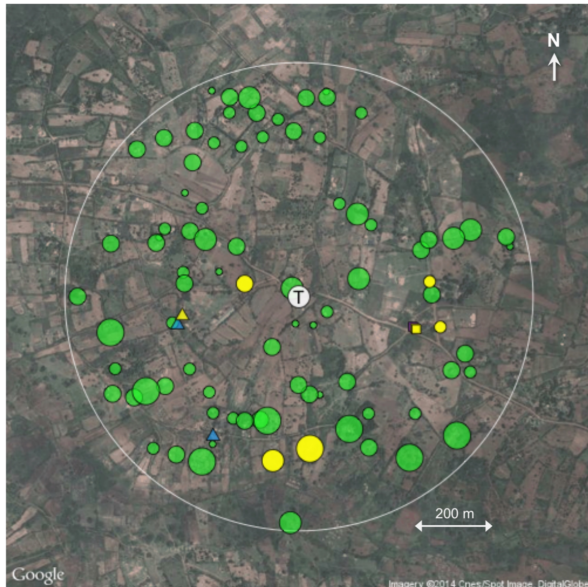
LMW (2019): Context



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	Study region	Nationwide county percentiles		
		25th	50th	75th
Total population	793,125	528,054	724,186	958,791
per square kilometer	375.4	39.5	183.2	332.9
% rural	85.7	71.6	79.5	84.4
% at school	44.7	37.0	42.4	45.2
% in school with secondary education	10.3	9.7	11.0	13.4
Total households	176,630	103,114	154,073	202,291
per square kilometer	83.6	7.9	44.3	78.7
% with high quality roof	59.7	49.2	78.5	88.2
% with high quality floor	27.7	20.6	29.7	40.0
% with high quality walls	32.2	20.3	28.0	41.7
% with piped water	6.3	6.9	14.2	30.6
Total public facilities	644	356	521	813
per capita (000s)	0.81	0.59	0.75	0.98
Electrification rates				
Rural (%)	2.3	1.5	3.1	5.3
Urban (%)	21.8	20.2	27.2	43.2
Public facilities (%)	84.1	79.9	88.1	92.6

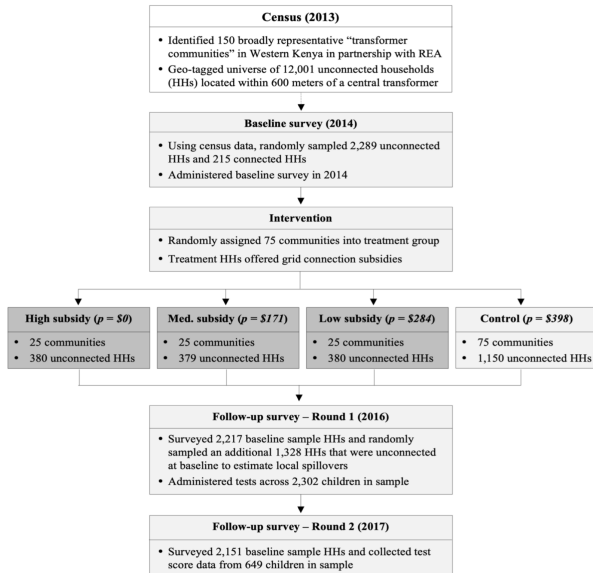
LMW (2019): Context



LMW (2019): Data

	Unconnected (1)	Connected (2)	<i>p</i> -value of diff. (3)
<i>Panel A: Household head (respondent) characteristics</i>			
Female (%)	62.9	58.6	0.22
Age (years)	52.3	55.8	< 0.01
Senior citizen (%)	27.5	32.6	0.11
Attended secondary schooling (%)	13.3	45.1	< 0.01
Married (%)	66.0	76.7	< 0.01
Not a farmer (%)	22.5	39.5	< 0.01
Employed (%)	36.1	47.0	< 0.01
Basic political awareness (%)	11.4	36.7	< 0.01
Has bank account (%)	18.3	60.9	< 0.01
Monthly earnings (USD)	16.9	50.6	< 0.01
<i>Panel B: Household characteristics</i>			
Number of members	5.2	5.3	0.76
Youth members (age ≤ 18)	3.0	2.6	0.01
High-quality walls (%)	16.0	80.0	< 0.01
Land (acres)	1.9	3.7	< 0.01
Distance to transformer (m)	356.5	350.9	0.58
Monthly (non-charcoal) energy (USD)	5.5	15.4	< 0.01
<i>Panel C: Household assets</i>			
Bednets	2.3	3.4	< 0.01
Sofa pieces	6.0	12.5	< 0.01
Chickens	7.0	14.3	< 0.01
Radios	0.35	0.62	< 0.01
Televisions	0.15	0.81	< 0.01
Sample size	2,289	215	

LMW (2019): Study design



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LMW (2019): Demand – Estimation

LMW randomly assigned subsidies:

- This means they can estimate the price elasticity simply

All they need to do is estimate:

$$Y_{ic} = \alpha + \tau_L D_c^L + \tau_M D_c^M + \tau_H D_c^H + \beta X_{ic} + \varepsilon_{ic}$$

where:

Y_{ic} is take-up for household i in community c

D_c^L , D_c^M , and D_c^H are treatment for the Low, Medium, and High subsidies

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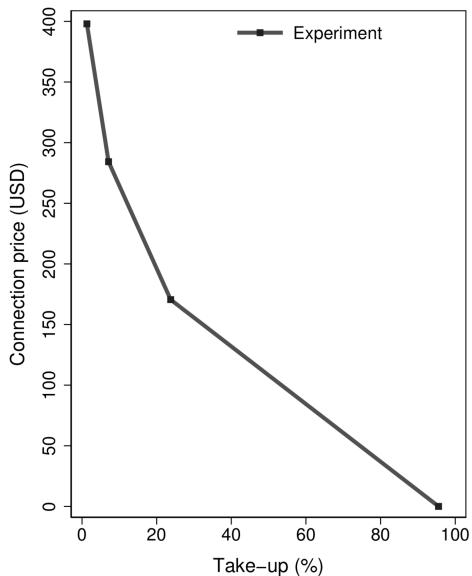
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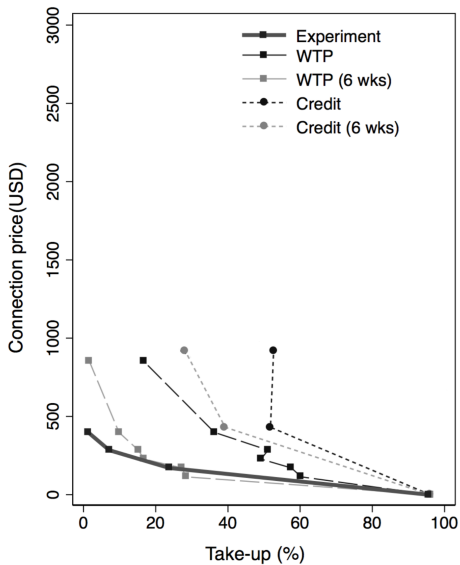
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- τ_L, τ_M, τ_H are (treatment) effects on take-up at different subsidy levels
- α captures take-up in the control group

LMW (2019): Demand – Results



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LMW (2019): Estimation – Supply

We need to know the cost of a connection:

- The average total cost (admin data) is \$1,813 per connection
- Are there “economies of scale”?
- To estimate this:

$$ATC_c = \tau_0 + \tau_1 Connections_c + \tau_2 Connections_c^2 + \varepsilon_c$$

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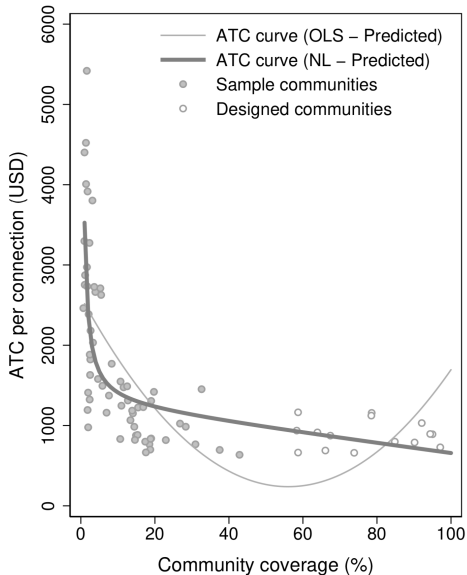
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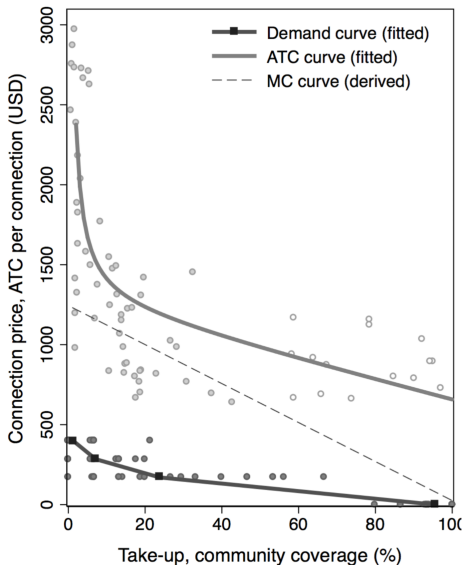
or

$$ATC_c = \underbrace{\frac{\tau_0}{\text{Connections}_c} + \tau_1}_{\text{fixed cost}} + \underbrace{\tau_2 \text{Connections}_c}_{\text{marginal cost}} + \varepsilon_c$$

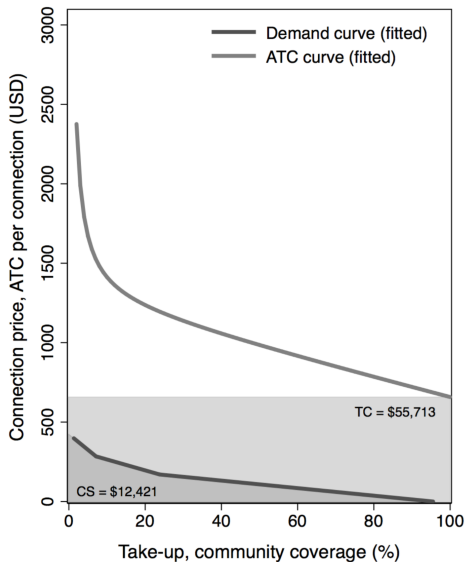
LMW (2019): Supply – Results



LMW (2019): Supply and demand – Results



LMW (2019): Supply and demand – Results



LMW (2019): Outcomes – Estimation

Following from the randomly-assigned subsidies:

- To get the ITT, they just estimate:

$$Y_{ic} = \alpha + \tau_H D_c^H + \beta X_{ic} + \varepsilon_{ic}$$

(no D_c^L or D_c^M because take-up was so low)

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- To get the ATT, they estimate:

$$E_{ic} = \alpha + \gamma_L D_c^L + \gamma_M D_c^M + \gamma_H D_c^H + \beta X_{ic} + \varepsilon_{ic}$$

and

$$Y_{ic} = \alpha + \tau \hat{E}_{ic} + \beta X_{ic} + \varepsilon_{ic}$$

→ They instrument to get from ITT to ATT

LMW (2019): Outcomes – Results

	Control (1)	ITT (2)	TOT (3)	FDR q -val (4)
<i>Panel A: Treatment effects on pre-specified outcomes</i>				
P1. Grid connected (%)	5.6 [23.0]	89.7*** (1.4)	–	–
P2. Monthly electricity spending (USD)	0.16 [1.29]	2.00*** (0.18)	2.20*** (0.20)	.001
P3. Household employed or own business (%)	36.8 [38.8]	5.1 (3.1)	4.5 (3.4)	.416
P4. Total hours worked last week	50.9 [32.8]	-2.8* (1.5)	-3.6** (1.7)	.167
P5. Total asset value (USD)	888 [851]	109 (108)	110 (120)	.540
P6. Ann. consumption of major food items (USD)	117 [92]	-3 (6)	-5 (7)	.548
P7. Recent health symptoms index	0 [1]	-0.03 (0.06)	-0.05 (0.07)	.548
P8. Normalized life satisfaction	0 [1]	0.12** (0.06)	0.13* (0.07)	.179
P9. Political and social awareness index	0 [1]	-0.03 (0.05)	-0.02 (0.05)	.731
P10. Average student test Z-score	0 [0.99]	-0.08 (0.10)	-0.10 (0.10)	.540
<i>Panel B: Mean treatment effects on grouped outcomes</i>				
G1. Economic Index (P3 to P6 outcomes)	0 [1]	0.06 (0.08)	0.03 (0.08)	–
G2. Non-Economic Index (P7 to P10 outcomes)	0 [1]	-0.01 (0.06)	-0.02 (0.07)	–

LMW (2019): Summary

	C	Experimental approach		Alternative approach		Key assumption(s)
		CS	NW	CS	NW	
Main estimates	658	147	-511	147	-511	
a) Income growth (<i>experimental approach</i>); Electricity consumption growth (<i>alternative approach</i>)	-	+139		-		Income growth of 3 percent per annum over 30 years (based on demand curves in figure 2, panel B); Electricity consumption growth of 10 percent per annum over 30 years (see table 4, column 2, row 3).
b) No credit constraints for grid connections	-	+301		-		Stated WTP without time constraints (see figure 5)
c) No transformer breakdowns	-	+33		+19		Reduce likelihood of transformer breakdowns from 5.4 to 0 percent (see appendix table B10).
d) No grid connection delays	-	+46		+26		Reduce waiting period from 188 to 0 days (see appendix figure A1).
e) No construction cost leakage	-140	-		-		Decrease total construction costs by 21.3 percent (see appendix table B8).
Ideal scenario	518	665	148	374	-144	